

Métrica inducida

Proposición 1 (Métrica inducida). *Sea V un espacio vectorial normado,*

$$\implies \forall x, y \in V : d(x, y) = \|x - y\|$$

define una métrica en V , llamada la métrica inducida por la norma.

Demostración: Basta con comprobar que se satisfacen las propiedades de la definición.

Sean $x, y, z \in V$, entonces:

- (i) $d(x, y) = \|x - y\| \geq 0$ y $d(x, y) = 0 \iff \|x - y\| = 0 \iff x = y$.
- (ii) $d(x, y) = \|x - y\| = \|(-1)(y - x)\| = |-1| \|y - x\| = \|y - x\| = d(y, x)$.
- (iii) $d(x, z) = \|x - z\| = \|(x - y) + (y - z)\| \leq \|x - y\| + \|y - z\| = d(x, y) + d(y, z)$. ■

Referenciado en

- `prop-desigualdad-triangular-inversa-norma`
- `teo-metrica-inducida-iff-homogeneidad-traslaciones`
- `esp-banach`

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